CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. (CE/IT/EC) Examination

November-December 2013

ME104 Basics of Mechanical Engineering

Date: 03.12.2013, Tuesday   Time: 10:00 a.m. To 01:00 p.m.   Maximum Marks: 70

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 (a) Answer the following MCQ's. [05]

1. The amount of heat require to raise the temperature of 1 kg of substance by 1°C is
   - a. Calorific value   b. Entropy
   - c. Specific heat   d. Enthalpy
2. Addition of sensible heat in water results in
   - a. Change in phase   b. Rise in temperature
   - c. Temperature and pressure   d. Change in phase and temperature
3. Steam generation rate is normally more in
   - a. Fire tube boiler   b. Water tube boiler
   - c. Vertical boiler   d. None of these
4. An economizer is used in a boiler to
   - a. Heat flue gases   b. Heat air for combustion
   - c. Heat feedwater   d. Superheat the steam
5. The COP of VCR system is normally
   - a. Less than 1   b. Equal to 1
   - c. More than 1   d. None of these

(b) Find out the Work done, Heat transfer and Change in Internal energy for Constant Pressure Process. [03]

OR

Explain Combined gas law. [03]

(c) Determine enthalpy and specific volume of steam under following conditions [04]

(i) 12.5 bar pressure and 0.9 dry
(ii) 12.5 bar pressure and dry saturated
(iii) 20 bar pressure and superheated of 300°C

Pressure P (bar)	Temperature Ts °C	H_f kJ/kg	H_{fg} kJ/kg	V_f m^3/kg	V_g m^3/kg
12.5	189.8	806.7	1977.4	0.001141	0.157
20	212.4	908.6	1888.6	0.001177	0.0995

(d) Define (i) Dryness Fraction (ii) Degree of superheat [02]

Q – 2 (a) A boiler generates 2500 kg of superheated steam per hour at a pressure of 11 bar and temperature at 200°C. The coal is supplied to the boiler at the rate of 300 kg/hr. The feed water temperature is 32°C and calorific value of coal is 31000 kJ/kg. Assume the specific heat of steam is 2.1 kJ/kg K. Calculate (i) boiler efficiency (ii) evaporation rate (iii) Generation factor (iv) Equivalent evaporation in kg of steam per hour. Assume $C_{pw} = 4.187$ kJ/kg K [06]

Pressure P (bar)	Temperature Ts °C	H_f kJ/kg	H_{fg} kJ/kg
11	184.1	781.1	1998.5

(b) Draw a neat sketch of Cochran Boiler showing all its components. Also explain its construction & working. [06]

OR

Explain working with neat sketch.

1. Water level indicator
2. Economizer [06]

Q-3 (a) Define unit of refrigeration and explain working of window air conditioner with neat sketch. [05]

(b) Define

1. Air conditioning and its applications
2. Coefficient of performance [04]

SECTION – II

Q-4 (a) Answer the following MCQ's. [05]

1. In a petrol engine, fuel and air is properly mixed in
   a. Fuel pump b. Cylinder
   c. Carburetor d. Spark plug
2. The suction in compression ignition engine contains
   a. Diesel only b. Petrol only
   c. Air only d. Air and diesel
3. Compressor is a machine which
   a. Provides power b. Delivers gas at low pressure
   c. Delivers gas at low temperature d. Delivers gas at high pressure
4. Automatic clutch normally used in two wheeler is known as
   a. Plate clutch b. Centrifugal clutch
   c. Jaw clutch d. Multiplate disc clutch
5. Reverted gear train is used when driver and driven shaft axis are
   a. Same axis b. Right angle axis
   c. Inclined axis d. Intersecting Axis

(b) Explain working processes of 4S compression ignition engine with neat sketch. [06]

OR

Explain working processes of 2S spark ignition engine with neat sketch. [06]

(c) A 4 cylinder 4S marine oil engine has cylinder diameter of 490 mm and piston stroke of 1000 mm. The engine uses 130 kg of fuel of calorific value 42000 kJ/kg per hour when running at 2 rev/s. The torque transmitted at the engine coupling is 22 kN.m and indicated mean effective pressure of 710 kN/m^2. Determine (i) indicated power (ii) brake power (iii) brake thermal efficiency (iv) mechanical efficiency (v) indicated thermal efficiency. [06]

Q - 5 (a) Write down classification of pumps and explain centrifugal pump and its components with neat sketch. [05]

(b) Write down classification of compressor and explain working of reciprocating compressor with net sketch. [04]

Q - 6 (a) Write classification of clutch and explain centrifugal clutch with neat sketch. [05]

OR

Write classification of coupling and explain Oldham coupling with neat sketch. [05]

(b) Write down the comparison between gear drive, chain drive and belt drive. [04]

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